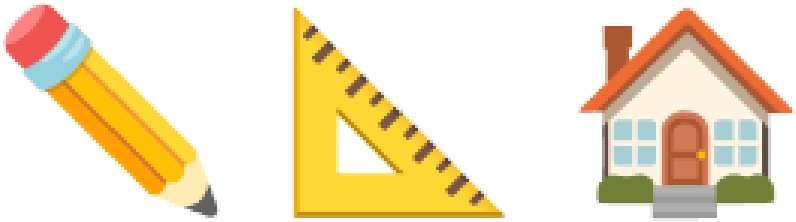
🏗️ Design a Sun Shelter — Part 1: Plan! 🏗️

Physical Science — Sunshine & Warmth — Unit 2, Lesson 5 of 6

📚 **Grade:** Kindergarten ⏱️ **Duration:** 30–45 minutes 🔬 **Subject:** Physical Science — Sunshine & Warmth
🔢 **Lesson:** 5 of 6



Every great building starts with a plan. Today we design — next lesson we build!

📋 **NGSS Standard:**
K-PS3-2 — Use tools and materials provided to design and build a structure that will reduce the warming effect of sunlight on an area

🎯 Learning Objective

Students will plan ideas for **"Design a Sun Shelter"** — Part 1: Plan, make observations during the activity, and explain their choices using evidence from the lesson.

🎯 The Big Idea

We've spent four lessons learning HOW sunlight warms surfaces. Now it's time to use that knowledge to **solve a real problem**! Scientists and engineers use what they know to design solutions. Today, students will think like engineers: identify the problem, gather ideas, choose materials, make a plan, and draw their design before they build it. *Planning before building leads to better results!*

🔧 The Engineering Design Challenge

"Design a shelter that keeps a small toy animal cool on a hot sunny day."

Available building materials for Lesson 6: paper, cardboard, aluminum foil, fabric scraps, tape

🔬 The Science Behind Good Design

Here are the ideas students have already explored in the earlier lessons:

- 🎨 **Color:** Light colors reflect sunlight; use white or silver (foil) on the outside
- 🌿 **Shade:** The bigger the shadow your shelter casts, the cooler the area underneath

> 💡 **Additional ideas to consider:** A pointed or sloped roof can help hot air rise, small openings can let trapped hot air escape, and thicker walls can slow heat transfer.

> 💡 **Key vocabulary:** design, engineer, shelter, reflect, insulate, ventilation, materials, plan, predict

📦 Materials for Planning Today

📝 What You Need for the Plan

- Student worksheet / design sheet (included)
- Pencils and crayons
- Ruler (optional, for drawing straight lines)

🏗️ Building Materials Available for Lesson 6

- Cardboard scraps (from cereal boxes, etc.)
- White and colored paper
- Aluminum foil
- Fabric scraps
- Tape (masking or clear)
- Scissors

🔬 Let's Get Started

Step 1 — Review What We've Learned (5 min)
Before designing, briefly review the key lessons:

- *"What color works best to stay cool?"* → Light colors or shiny silver (reflects light)
- *"Does shade help?"* → Yes! Blocking direct sunlight is the most important thing
- *"Did foil keep the ice cool in Lesson 4?"* → Yes! Because it reflects sunlight

Use this knowledge in your design!

Step 2 — Brainstorm (5–10 min)
Ask: *"What shape should the shelter be? What color? What should it be made of?"*
Encourage them to think about:

- A roof that slopes (sheds heat, like a real house)
- Light colors or foil on the outside surface
- A design that would shade the toy animal completely

Let them sketch rough ideas first — even just shapes and colors.

Step 3 — Create the Design Plan (15–20 min)
Students draw their shelter design on the worksheet. They should:

- Draw the shelter from the side (side view)
- Draw the toy animal inside it
- Label or color the materials they'll use
- Check off which building materials they want to use
- Write or dictate a prediction: *"My shelter will work because ___"*

Step 4 — Share & Discuss (5 min)
Have each student share their design. Ask:
"Why did you choose that color? Why that shape?"
Help them make connections to the science: *"Your foil roof will reflect sunlight — just like foil kept ice cool in our melting race!"*

💬 Discussion Question

🗣️ **"A white roof and a black roof are both in the sun. Which house do you think is cooler inside?"**

> *→ The white roof house! White reflects sunlight while black absorbs it, turning that light energy into heat that enters the building. That's why many buildings in hot climates (like in Greece, Spain, and the Middle East) have brilliant white roofs! Green roofs covered in plants work well too — plants use sunlight for photosynthesis instead of turning it all into heat, and water in the plants evaporates, cooling the area (just like our wet cloth in Lesson 1).*

🚀 Extensions

- 🌍 **Real Architecture:** Look up photos of houses in Santorini, Greece (white), or desert adobe homes (thick clay walls). How do their designs keep people cool?
- 🌿 **Green Roofs:** Search for "green roof buildings" — some city buildings have gardens on their roofs! Why would plants help keep buildings cool?
- 🏜️ **Ancient Wisdom:** Ancient people in hot deserts built thick-walled mud brick homes to stay cool. Why do thick walls help? (They slow heat transfer from outside to inside.)
- ✏️ **Draw More Designs:** Can you design two different shelters using different approaches? Which do you think would work better?

👩‍🏫 Parent/Teacher Notes

- **This is a 2-part lesson:** Lesson 5 = Plan, Lesson 6 = Build & Test. Today's goal is fully fleshed-out design plans, not a complete shelter.
- **Encourage creativity:** There's no single right answer! A child who designs a white-roofed box with foil walls is applying science. So is one who designs a pointed tent-shape from white paper.
- **Science-based reasoning:** Gently push kids to connect their design choices to what they learned. "Why white?" "Why that shape?" The reasoning is as important as the design itself.
- **Save the plans:** They'll refer back to them in Lesson 6 when they actually build!
- **Assessment:** Can the child explain at least one design choice using science? (e.g., "I'm using foil because it reflects sun") That's K-PS3-2 in action.

🏗️ Student Worksheet

Name: _____ Date: _____

✏️ Draw Your Shelter Design:

Draw your shelter from the *side*. Show the toy animal inside. Color and label the materials you'll use!

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Draw your shelter design here!

Show the toy animal inside 🧸

🧰 Materials I Plan to Use:

Check all the materials you want to use to build your shelter:

- ☐ 📦 Cardboard
- ☐ 📄 White Paper
- ☐ 🪩 Aluminum Foil
- ☐ 🧶 Fabric Scraps
- ☐ 🖍️ Colored Paper
- ☐ 🖊️ Tape

🔬 My Prediction:

My shelter will keep the animal cool because:

_____

The most important part of my design is:

🔑 PARENT/TEACHER ANSWER KEY — Reference Guide

What Makes a Strong Design?

There's no single "correct" design — but look for these science-based reasoning elements:

Design Feature	Why It Works (Science)	Materials That Help
Light/white outer surface	Reflects sunlight, absorbs less heat	White paper, foil
Sloped or pointed roof	Sheds heat, lets hot air rise and escape	Any material shaped into a slope
Foil lining	Reflects radiant heat, great insulator	Aluminum foil
Good overall coverage	Blocks all direct sunlight from reaching animal	Cardboard, fabric, paper
Ventilation gaps	Trapped hot air escapes, cool air enters	Leave small openings in design

💡 Engineering Notes for Parents

The engineering design process has 5 stages: **Ask → Imagine → Plan → Create → Improve**. This lesson covers Ask, Imagine, and Plan. Lesson 6 covers Create and Improve. Expect designs to be imperfect — that's normal and valuable! The real learning happens when they test in Lesson 6 and have to improve their design.

✅ What Success Looks Like

- Child produces a drawing of a shelter with the toy animal inside
- Child can explain at least one design choice using science from previous lessons
- Child has identified which materials they'll use
- Child makes a testable prediction: "My shelter will work because..."

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